

Robotics

A robot is a type of smart, automated machine that can be programmed to perform different tasks, often with little or no supervision. Robots now perform a wide range of tasks, from cleaning skyscraper windows to assisting with surgical operations, sometimes with more accuracy or greater force than humans can manage. Robots are often found performing work that humans find unpleasant, repetitive, or impossible.

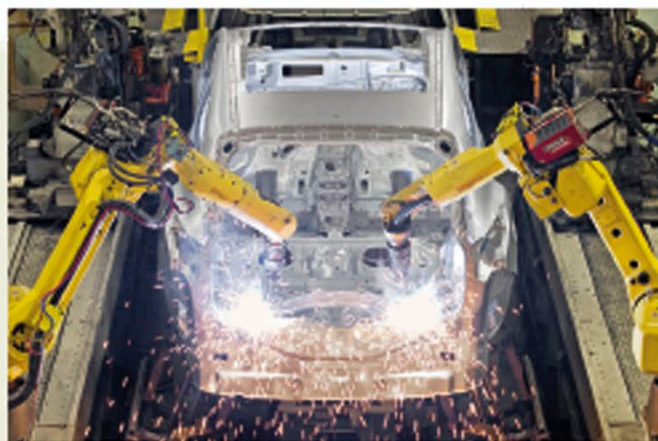
Military robots

Robots can make excellent security guards or roam ahead of human forces as expendable spies or scouts. Some seek out survivors in disaster areas, while others investigate danger zones such as minefields or toxic chemical spills.

Cameras send detailed views to the human bomb disposal team.

A RONS robot, designed by a company called Remotec, disposes of an unexploded bomb.

Robot uses a sensitive gripper to handle an unexploded shell.

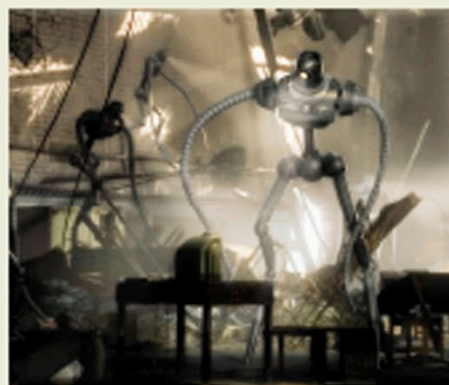


Working in factories

The first robot that worked in a factory was a Unimate, which handled red-hot metal castings at an American car-making factory in 1961. There are now more than 1.5 million robots in factories assembling products, welding, picking, and packing objects with pinpoint precision, and spray-painting products.

Robots in fiction

Robots emerged in science fiction before they existed in the real world. They are often portrayed as highly intelligent, thinking machines. In reality, robots need to be programmed by humans, although some can learn from their surroundings.



Robot from the science fiction film *Sky Captain and the World of Tomorrow*

Key events

1921

Czech playwright Karel Čapek popularized the term "robot" in a play called *R.U.R.* The term comes from the Czech *robota*, which means drudgery, or forced labor.

1966

Shakey became the first mobile robot able to navigate its way around a series of rooms, using cameras and sensors, in California.

1975

A six-jointed, electric robot arm called Programmable Universal Manipulation Arm (PUMA) was devised by American inventor Victor Scheinman. It proved influential in industrial robot design.

1997

The Sojourner rover was the first robot to move around another planet (Mars) on its electric motor-driven wheels. It sent images from its cameras back to Earth.

Shakey

